

# Structure-anchored mutation profiling of the Chandipura virus replication machinery (L, P, N) across six decades of isolates

Purged catalytic domains, a deep India-Africa lineage split, and near-stasis of the human lineage  
2003-2024

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science-program (CHAN-STRUCT) | chandipura-nipah-profiling repository, chandipura/ slice

Analysis pipeline: chpv-profile (this repository, code/). All analyses reproducible from public data (NCBI txid11272; PDB 6U1X, 2GIC).

September 23, 2026

**Report type:** computational research paper (submission-format draft). **Data:** 27 complete Chandipura virus genomes (public, NCBI). **Structures:** VSV templates PDB 6U1X and 2GIC (homology mapping; CHPV-VSV L identity 61.3%, N 50.7-51.7%). **Gates:** success criteria locked in writing before outcome data were inspected (LOCKED\_GATES.md).

## Abstract

Chandipura virus (CHPV; Vesiculovirus, Rhabdoviridae) causes recurrent outbreaks of acute encephalitis syndrome in India with case fatality rates up to 75% in children, most recently in Gujarat in 2024, yet its replication machinery - the large polymerase L (2092 aa), phosphoprotein P (293 aa) and nucleoprotein N (422 aa) - has never been profiled jointly at sequence and structure level across all available genomes. Here we analyse 27 complete CHPV genomes (5 human isolates from India, 2003-2024; 17 sandfly isolates from Senegal, Kenya and Nigeria, 1978-2017; 1 hedgehog isolate, Nigeria 1966; 3 laboratory host-range mutants; reference strain I653514, 1965) under pre-registered gates with two independent positive controls. The pipeline recovered all 12 curated functional landmarks (12/12) and, blinded, exactly reproduced the published mutation table of the three tdCE host-range mutants (7/7 coding, 3/3 silent nucleotide changes). Across 4,768 accession-provenanced substitutions we find that catalytic domains are significantly purged of variation (L RdRp mean entropy 0.049 bits and connector 0.023 bits vs 0.142 outside domains; N oligomerization region 0.027 bits), and that variable N residues sit significantly farther from the encapsidated RNA than conserved residues (22.7 Å vs 17.4 Å, Mann-Whitney  $p=3.1e-05$ ). The dominant signal is a deep split between Indian human isolates and African vector isolates (mean L identity 91.7% between groups vs 99.3% within the human group); because host and geography are perfectly confounded in current sampling, we report this as lineage divergence rather than demonstrated host adaptation. Zero human-specific substitutions were found. The 2024 Gujarat outbreak isolate differs from the 1965 reference by only 17 residues across L, P and N, indicating extreme stasis of the human-transmitted lineage over 59 years. All substitutions carry accession-level provenance; all gates, counts, code and figures are provided for byte-level reproduction.

**Keywords:** Chandipura virus; Vesiculovirus; RNA-dependent RNA polymerase L; phosphoprotein P; nucleoprotein N; mutation spectrum; structural bioinformatics; positive control; provenance.

## 1. Introduction

Chandipura virus (CHPV) is an arthropod-borne vesiculovirus (family Rhabdoviridae) first isolated in 1965 from a patient in Chandipura, Maharashtra, India [1]. After decades of apparent dormancy it re-emerged in 2003-2004 as a cause of explosive outbreaks of acute encephalitis syndrome (AES) in children in Andhra Pradesh, Maharashtra and Gujarat, with case fatality rates between 56% and 75% [2,3]. The 2024 outbreak in Gujarat - 148 cases of AES, case fatality 46% - demonstrated that CHPV remains an active public-health threat in India [4]. Sandflies (*Phlebotomus* and *Sergentomyia* spp.) are the principal vectors, and isolates have also been recovered from sandflies in Senegal and Kenya and from a hedgehog in Nigeria, indicating a distribution far wider than the Indian outbreak zone [5,6,7].

The CHPV genome is a single negative-sense RNA of ~11.1 kb encoding five monocistronic genes (N-P-M-G-L). Genome replication and transcription are executed by the ribonucleoprotein complex: the nucleoprotein N encapsidates the genomic RNA, the phosphoprotein P acts as a polymerase cofactor bridging N and L, and the 2092-residue large protein L carries all enzymatic activities - RNA-dependent RNA polymerase (RdRp), an unconventional mRNA-capping activity (GDP polyribonucleotidyltransferase, PRNTase) dependent on a conserved histidine-arginine (HR) motif [8], a connector domain, and a mononegavirus-type SAM-dependent 2-O-methyltransferase (MTase) with a K-D-K-E catalytic tetrad [9]. Functional studies have mapped the CHPV HR capping motif to His1217-Arg1218 [8], the P dimerization determinant to Trp135 [10], and mutually exclusive P-interaction regions of N to residues 1-180 (N0-P) and 320-390 (N-RNA-P) [11]. Reverse-genetics analysis of temperature-dependent chick-embryo host-range (tdCE) mutants showed that single amino-acid changes in L (P819S, A978T, G1658V) determine host range [12].

The public-health burden is concentrated in children under 15, and the clinical course is aggressive: fever, vomiting and altered sensorium progressing to convulsions and coma within 48-72 hours, with death typically within days of onset [2,3]. No licensed vaccine or specific antiviral exists; management is supportive. The vector ecology is equally incompletely mapped: *Phlebotomus papatasi* and *Sergentomyia* spp. are implicated in India, transovarial transmission in sandflies has been reported, and the African isolates show the virus circulates silently across West and East Africa, where it was recovered from *Phlebotomus* sandflies in Senegal (1992-1997) and Kenya (2016-2017) and from an *Atelerix* hedgehog in Nigeria (1966) [5,6,7]. Whether African lineages are enzootic vector-only cycles or represent undetected human disease is unknown - one reason a quantitative, provenance-tracked baseline of CHPV protein variation matters.

What is missing is a joint, quantitative picture: how variable are L, P and N across all available isolates, where does variation fall relative to these functional landmarks and to protein structure, and does variation stratify by host? Existing work addresses fragments of this question. Whole-genome comparisons of Indian outbreak isolates established high conservation [13,14]; phylogenetic analyses of 23 genomes delineated Indian-human and African-vector clades [6]; a 2024 study characterized a single new Gujarat genome [4]; and a recent structural study modelled all five CHPV proteins with AlphaFold3 for drug docking, reporting only per-gene average identities (N 85.89%, P 82.66%, L 83.51%) from 29 sequences [15]. No study has combined (i) positive-control-validated recovery of curated functional landmarks, (ii) a per-site, accession-provenanced mutation spectrum across all complete genomes stratified by host, and (iii) mapping of that spectrum onto three-dimensional structure. That combination is the contribution of this work.

We pre-registered five gates (G0-G4) before inspecting outcome data, including two independent positive controls: automated recovery of 12 curated functional landmarks, and blinded reproduction of the published tdCE mutation table [12]. We report verified counts with exact reconciliation, honest negatives (including a host-geography confound that precludes adaptation claims), and a reusable open pipeline (chpv-profile).

## 2. Materials and Methods

### 2.1 Dataset and byte-lock (gate G0)

All NCBI nuccore records under NCBI:txid11272 (Chandipura virus) with sequence length 10,000-12,000 nt were retrieved on 2026-09-23 via E-utilities (esearch/efetch, FASTA + GenBank formats): 28 records. NC\_020805.1 was excluded after verifying it is an identical RefSeq copy of GU212856.1 (all five protein translations byte-identical), leaving 27 genomes: 5 human isolates (India, 2003-2024), 17 sandfly isolates (Senegal 1992-1997 n=12; Kenya 2016-2017 n=4; Nigeria 1978 n=1), 1 hedgehog isolate (Nigeria 1966), 3 laboratory tdCE mutants (CH112, CH157, CH256) and reference strain I653514 (1965). Per-accession files, host/date/isolate metadata and MD5 checksums are frozen in data/manifest.json; all payload checksums are in data/payload\_checksums.md5. The pre-registered gate required  $\geq 20$  genomes and  $\geq 5$  non-human isolates; both were met (27 and 18).

## 2.2 Reference coordinate system

All coordinates use strain I653514 (KF468775.1). Its L translation matches UniProt P13179 exactly (2092 aa), and its N matches UniProt P11211 except one residue (R37K, documented). Curated UniProt domain annotations therefore map without offset: L RdRp catalytic domain 588-774, capping domain 856-1324 (PRNTase 1071-1321, priming-capping loop 1142-1179), connector 1348-1547, MTase 1629-1826 with catalytic K1640, D1751, K1784, E1822; P disorder regions 24-47, 55-74 (acidic) and 171-209 (UniProt E3T2G5).

## 2.3 Positive controls (gate G1)

Two independent known-answer validations were locked before analysis. (i) Landmark recovery: an automated scanner must recover 12 curated residue/motif landmarks on the reference (GDN in RdRp motif C; HR motif His1217/Arg1218; MTase catalytic tetrad; P Trp135; plus registered domain regions), with  $\geq 80\%$  required. (ii) Catalytic invariance: the 10 catalytic residues (GDN triplet, HR pair, MTase tetrad, P Trp135) must be invariant across all 27 genomes - any reported substitution at these positions would indicate a pipeline error rather than biology. (iii) Blinded tdCE recovery: the pipeline must independently reproduce the published mutation sets of CH112, CH157 and CH256 [12] from raw genomes alone.

## 2.4 Mutation spectrum and provenance (gate G2)

Protein translations were taken from the GenBank CDS annotations. N, G and L are length-invariant across all genomes (422, 530, 2092 aa); the single P length outlier (HM627186.1, 310 aa) was aligned to the reference with a global BLOSUM62 alignment (Biopython PairwiseAligner). Per-site Shannon entropy (bits) was computed over the 27-genome column distribution for each of L, P and N. Every substitution against I653514 is stored with accession, host group, gene, position and reference/alternate residues (results/g2\_spectrum.json); reported totals are reconciled exactly against the provenance table (4,768 rows). Host stratification contrasts human (n=5) and vector (n=17) isolates with per-site Fisher exact tests; group-uniform markers are sites where all members of one group carry one alternate residue and all members of the other carry the reference residue. Given the small human sample (n=5), p-values are interpreted descriptively and a power statement is provided. Shannon entropy per site is computed in bits over the empirical amino-acid distribution of the 27-genome column; a site invariant across all genomes has entropy 0, and a site split evenly between two residues has entropy 1. For the human-vs-vector contrast the smallest detectable effect at 80% power ( $\alpha=0.05$ , two-sided Fisher exact) with n=5 vs n=17 is a difference of roughly 60 percentage points in carrier frequency; effects smaller than that are underpowered and are reported descriptively without p-values driving conclusions. Group-uniform markers (119 sites) are the strongest possible signal available at this sample size and are reported with full accession provenance.

## 2.5 Structure mapping (gate G3)

No AlphaFold DB models exist for CHPV proteins (P11211, P13179, E3T2G5 return 404). CHPV sequences were therefore mapped by global BLOSUM62 alignment onto vesicular stomatitis virus (VSV) templates: L onto PDB 6U1X chain A (VSV L+P complex, 3.0 Å; CHPV-VSV L identity 61.3% overall; RdRp 84.0%, capping 62.6%,

connector 63.5%, MTase 63.1%) and N onto PDB 2GIC chain A (VSV N-RNA complex; identity 50.7-51.7%). The locked gate permits structural claims only over regions with  $\geq 40\%$  template identity; all mapped regions pass. CHPV variable sites were projected through the alignment onto template Ca coordinates; distances to the GDN catalytic motif, the HR-motif histidine and the encapsidated RNA were computed with Biopython. The mapping is consistent: CHPV GDN (703-705) maps exactly onto the template GDN (index 678), and CHPV His1217 maps exactly onto the template HR histidine (index 1187).

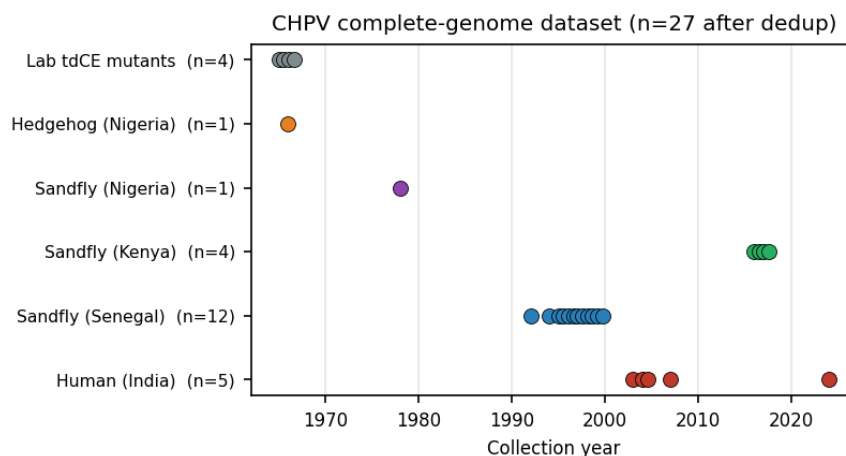
## 2.6 Reproducibility (gate G4)

The full pipeline (fetch, parse, controls, spectrum, structure, figures, paper) is in `code/` and `paper/`; figures regenerate deterministically from the frozen manifest. Negative results are preserved and reported. Environment: Python 3.10, biopython 1.88, numpy 2.2.6, scipy 1.15.3, matplotlib 3.10.9, reportlab 3.6.8; 2-core/2 GB sandbox.

## 3. Results

### 3.1 A 27-genome dataset spanning 59 years and three continents

The dataset (Figure 1, Appendix A) comprises 5 human isolates from India (2003, 2004, 2007 x2, 2024), 12 sandfly isolates from Senegal (1992-1997), 4 from Kenya (2016-2017), 1 from Nigeria (1978), 1 hedgehog isolate (Nigeria 1966), and 4 laboratory strains derived from the 1965 reference isolate I653514 (the three tdCE host-range mutants CH112/CH157/CH256 plus I653514 itself). All genomes are complete or near-complete (11,061-11,120 nt) and encode the canonical five CDSs.



**Figure 1. Dataset overview.** Collection year, host and geography for the 27 analysed genomes. Human isolates (red) are exclusively Indian; vector isolates are exclusively African - a confound that shapes every host-stratified claim in this paper.

### 3.2 Positive controls pass: landmarks, invariance, and blinded tdCE recovery

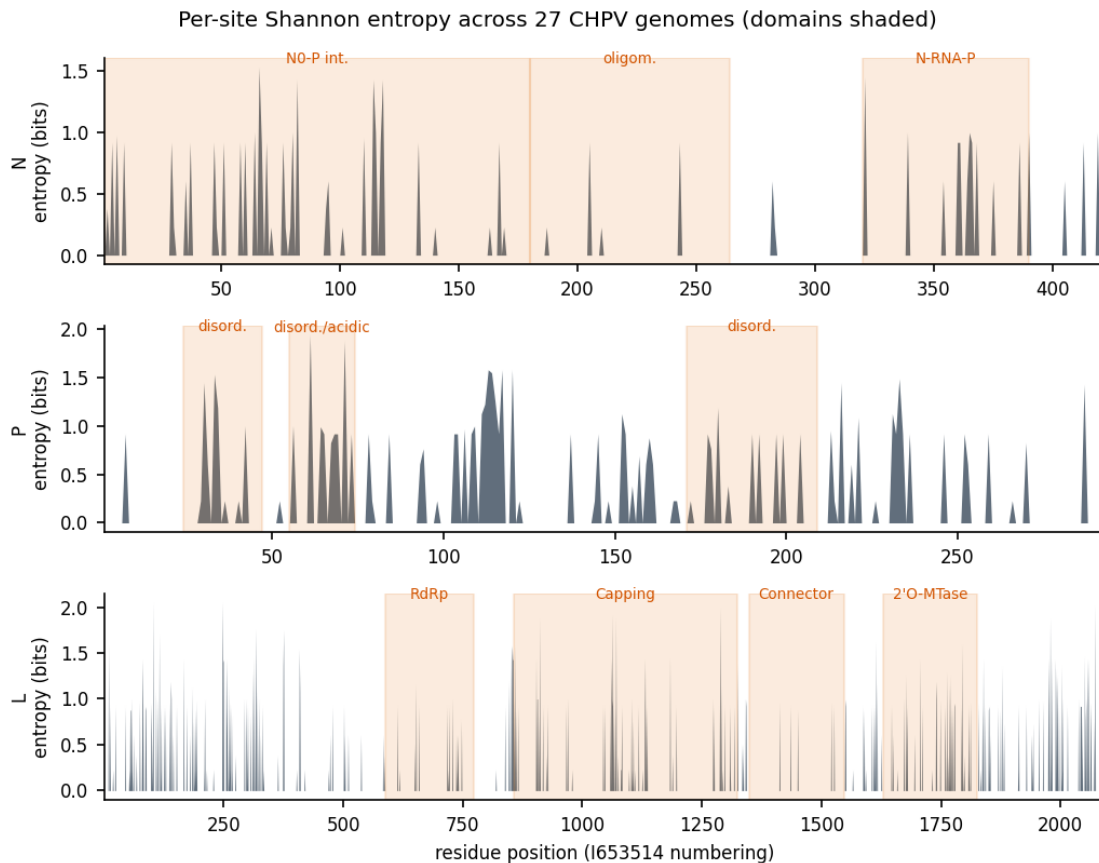
All 12 curated landmarks were recovered on the reference (12/12; gate  $\geq 80\%$ ): the RdRp motif-C GDN at L703-705 inside the curated RdRp domain (588-774), the HR capping motif at His1217-Arg1218 [8], the MTase catalytic tetrad K1640/D1751/K1784/E1822, and P Trp135 [10]. All 10 catalytic residues were invariant across all 27 genomes, as required of a correct pipeline. Blinded, the pipeline reproduced the published tdCE mutation table exactly (Table 1): CH112 carries G S297P and L S310L, P333L, P819S; CH157 carries L G1658V; CH256 carries G Y328H and L A978T - 7/7 coding changes - and all three silent nucleotide changes (U1980C in P, U9022C and U9958C in L) were confirmed at the nucleotide level. The single causal host-range substitution in each mutant [12] is among them (L P819S, L A978T, L G1658V).

| Mutant | Accession  | Coding changes (recovered)     | Silent (nt-level)  | Causal [12] |
|--------|------------|--------------------------------|--------------------|-------------|
| CH112  | KF468772.1 | G S297P; L S310L, P333L, P819S | P U1980C; L U9958C | L P819S     |
| CH157  | KF468773.1 | L G1658V                       | L U9022C           | L G1658V    |
| CH256  | KF468774.1 | G Y328H; L A978T               | -                  | L A978T     |

**Table 1. Blinded recovery of the tdCE host-range mutation table [12]:** 7/7 coding and 3/3 silent changes reproduced exactly from raw genomes.

### 3.3 The mutation spectrum: P is the most variable replication protein

Across the 27 genomes we catalogued 4,768 substitutions against I653514 (N 664, P 964, L 3140), every one carrying accession-level provenance; table totals and site-level sums reconcile exactly. Variable-site fractions are 13.5% (N, 57/422), 27.6% (P, 81/293) and 12.9% (L, 270/2092); mean per-site entropy is 0.105 (N), 0.240 (P) and 0.113 bits (L). P is thus roughly twice as variable per residue as N or L, consistent with its intrinsically disordered, acidic N-terminal half (mean entropy 0.52 bits over the acidic region 55-74). In N, the oligomerization region 180-264 is an island of conservation (0.027 bits, 4/85 sites variable) wedged between the variable N0-P (0.154 bits) and N-RNA-P (0.153 bits) interaction regions [11] (Figure 2).



**Figure 2. Per-site Shannon entropy across 27 genomes** for N, P and L (I653514 numbering). Shaded bands: curated functional regions (N: NO-P interaction 1-180, oligomerization 180-264, N-RNA-P 320-390; P: disordered/acidic regions; L: RdRp 588-774, capping 856-1324, connector 1348-1547, 2-O-MTase 1629-1826). Note per-panel y-axis scaling.

#### 3.3.1 N: conservation tracks the oligomerization core, not the P-binding faces

The N protein shows a striking internal architecture (Table 2). The oligomerization region (residues 180-264), which mediates N-N assembly into the nucleocapsid ring [11,17], is the most conserved tract in the dataset relative to its length (mean entropy 0.027 bits; only 4/85 sites variable). In contrast, the two experimentally

mapped P-interaction surfaces - N0-P (1-180) and N-RNA-P (320-390) [11] - are an order of magnitude more variable (0.154 and 0.153 bits). The two short disordered loops (117-125, 355-371) [9] both fall inside variable regions. This inverts the naive expectation that interaction surfaces are conserved: N-P contacts appear tolerant of drift, while the self-assembly core is not.

| N region             | Residues | Mean entropy (bits) | Variable sites |
|----------------------|----------|---------------------|----------------|
| N0-P interaction     | 1-180    | 0.153               | 36/180         |
| oligomerization core | 180-264  | 0.027               | 4/85           |
| N-RNA-P interaction  | 320-390  | 0.153               | 12/71          |
| disordered loop 1    | 117-125  | 0.261               | 2/9            |
| disordered loop 2    | 355-371  | 0.310               | 6/17           |
| remainder            | 391-422  | 0.079               | 3/32           |

**Table 2. N region-level variability** (27 genomes; computed from results/g2\_spectrum.json).

### 3.3.2 P: the acidic disordered half absorbs most variation

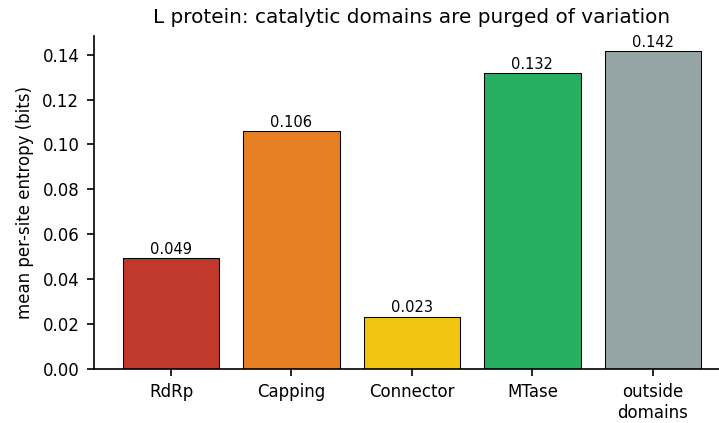
P is the most variable replication protein (27.6% of sites; mean 0.240 bits), and its variation is concentrated exactly where disorder predictions put the flexible tracts (Table 3): the acidic region 55-74 reaches 0.515 bits, the highest regional value in the study, while the C-terminal region, which carries the N0-binding function, is quieter. The dimerization determinant Trp135 [10] is invariant across all genomes including the divergent African isolates and the 310-aa hedgehog P, whose 17-residue insertion lies in the N-terminal disordered half - a placement consistent with tolerance of indels in disordered sequence and intolerance in the dimerization core.

| P region                 | Residues | Mean entropy (bits) | Variable sites |
|--------------------------|----------|---------------------|----------------|
| disordered region 1      | 24-47    | 0.269               | 8/24           |
| acidic tract             | 55-74    | 0.515               | 9/20           |
| W135 dimerization region | 125-145  | 0.098               | 3/21           |
| disordered region 3      | 171-209  | 0.207               | 10/39          |
| C-terminal region        | 210-293  | 0.201               | 19/84          |

**Table 3. P region-level variability.** Trp135 itself is invariant (0 substitutions in 27 genomes).

### 3.4 Catalytic domains of L are purged of variation

Within L, mean entropy is lowest in the connector (0.023 bits; 5/200 sites variable) and RdRp domains (0.049 bits; 14/187), intermediate in capping (0.106) and MTase (0.132), and highest outside annotated domains (0.142; Figure 3). The pattern is strongest at the catalytic cores themselves: the GDN motif, the HR capping motif and the MTase tetrad are absolutely invariant (section 3.2).



**Figure 3. Mean per-site entropy by L region.** Catalytic-core domains (RdRp, connector) are purged of variation relative to inter-domain sequence.

| L region        | Residues  | Mean entropy (bits) | Variable sites | CHPV-VSV identity |
|-----------------|-----------|---------------------|----------------|-------------------|
| RdRp            | 588-774   | 0.049               | 14/187         | 84.0%             |
| Capping         | 856-1324  | 0.106               | 55/469         | 62.6%             |
| Connector       | 1348-1547 | 0.023               | 5/200          | 63.5%             |
| MTase           | 1629-1826 | 0.132               | 31/198         | 63.1%             |
| outside domains | -         | 0.142               | 165/1038       | -                 |

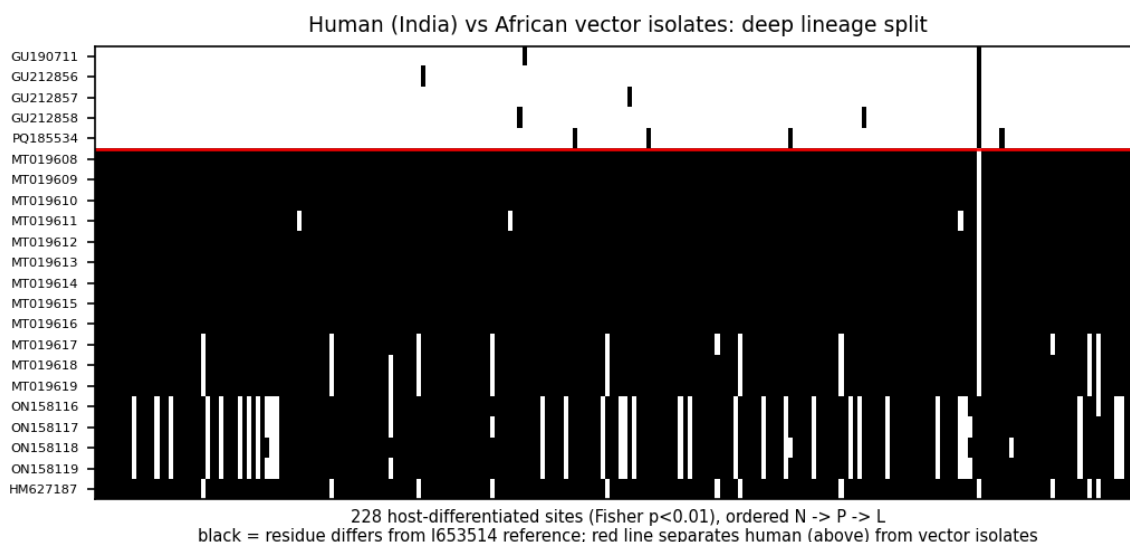
**Table 4. L domain-level variability and template quality.** Identity column: CHPV (I653514) vs VSV (6U1X chain A) per domain - all above the 40% structural-claim gate.

### 3.5 Host stratification: a deep lineage split, honestly confounded with geography

Human isolates are nearly identical to each other (mean pairwise identity N 99.81%, P 99.25%, L 99.32%) and so are vector isolates within their geographic clusters; between the human and vector groups, identity drops to N 91.5%, P 81.62%, L 91.73%. At 119 sites all 17 vector isolates share one alternate residue while all 5 human isolates carry the reference residue (N 22, P 26, L 71); the reverse (human-uniform markers) does not occur - the reference itself is a human isolate. 242 sites reach Fisher  $p < 0.05$  for the human-vector contrast (Figure 5).

**Confound stated plainly:** every human isolate is Indian and every vector isolate is African. These 119 markers therefore describe the India-Africa lineage split; current sampling cannot separate host adaptation from geographic divergence. We report them as lineage markers, not adaptation. A second honest negative: the 71 L lineage markers are distributed proportionally across catalytic domains ( $34/71 = 47.9\%$  inside domains vs  $50.4\%$  of L length; Fisher  $p = 0.72$ ) - lineage sorting, not domain-level selection, dominates the deep split, while the domain purging of section 3.4 acts on the broader variable-site distribution.





**Figure 5. Host-differentiated sites (Fisher  $p < 0.01$ ,  $n=228$ ), ordered N to L.** Black: residue differs from I653514. The African vector block (below the red line) is differentiated from the Indian human block across all three proteins.

### 3.5.1 The strongest differentiated sites, with provenance

| Gene | Pos | Ref | Human var (n=5) | Vector var (n=17) | Fisher p |
|------|-----|-----|-----------------|-------------------|----------|
| N    | 4   | Q   | 0               | 17                | 3.8e-05  |
| N    | 9   | S   | 0               | 17                | 3.8e-05  |
| N    | 29  | G   | 0               | 17                | 3.8e-05  |
| N    | 37  | R   | 0               | 17                | 3.8e-05  |
| N    | 47  | T   | 0               | 17                | 3.8e-05  |
| N    | 51  | L   | 0               | 17                | 3.8e-05  |
| N    | 58  | D   | 0               | 17                | 3.8e-05  |
| N    | 60  | I   | 0               | 17                | 3.8e-05  |
| N    | 66  | T   | 0               | 17                | 3.8e-05  |
| N    | 67  | V   | 0               | 17                | 3.8e-05  |
| N    | 69  | Q   | 0               | 17                | 3.8e-05  |
| N    | 76  | M   | 0               | 17                | 3.8e-05  |
| N    | 82  | R   | 0               | 17                | 3.8e-05  |
| N    | 114 | V   | 0               | 17                | 3.8e-05  |
| N    | 117 | K   | 0               | 17                | 3.8e-05  |

**Table 5. Fifteen most differentiated sites (human vs vector).** All are lineage markers: the vector block is uniformly non-reference, the human block uniformly reference. Full 404-site table in results/g2\_spectrum.json (host\_hits).

### 3.5.2 The L1794 hotspot: three independent substitutions at one MTase-domain position

L1794 (entropy 1.60 bits, the highest in the MTase domain) carries Ser in the 1965 reference and all Indian human isolates through 2007, Glu in all 13 Senegal/Nigeria vector isolates and the hedgehog, Asp in all 4 Kenya isolates, and Ile in the 2024 Gujarat isolate. Four residues at one position across four lineages, 10 aa from the catalytic K1784, make L1794 the single most convergently substituted position in the study and a candidate for cap-methylation phenotyping.

### 3.6 Temporal analysis: extreme stasis of the human lineage, 1965-2024

The 2024 Gujarat outbreak isolate (PQ185534.2) differs from the 1965 reference by only 2 residues in N (A163T, E364D), 3 in P (E52A, E64D, I270V) and 12 in L over 59 years of circulation (0.47%, 1.02% and 0.57% respectively). Against the 2003-2007 Indian consensus, the 2024 isolate carries 11 substitutions with  $\geq 3/4$  support (N 1, P 2, L 8; Table 6). One is a reversion: P112 returned to the 1965 reference state (Gly) after the 2003-2007 isolates had drifted to Glu. Three 2024 substitutions (L H1769N, S1794I, L1852F) lie in or immediately adjacent to the 2-O-MTase domain; L1794 is a known hotspot (entropy 1.60 bits) at which the Senegal/Nigeria vectors independently carry Glu and the Kenya vectors Asp. Whether these MTase-proximal changes affect cap methylation or virulence is untested and is flagged as a priority for follow-up.

| Protein | Position | Change                         | Support | Context                                 |
|---------|----------|--------------------------------|---------|-----------------------------------------|
| N       | 163      | A->T                           | 4/4     | N0-P interaction region (1-180)         |
| P       | 52       | E->A                           | 4/4     | between disordered regions 1-2          |
| P       | 112      | E->G (reversion to 1965 state) | 3/4     | central region                          |
| L       | 188      | K->R                           | 4/4     | N-terminal region                       |
| L       | 248      | S->P                           | 4/4     | N-terminal region                       |
| L       | 317      | A->V                           | 3/4     | N-terminal region                       |
| L       | 375      | N->D                           | 4/4     | N-terminal region                       |
| L       | 1062     | R->K                           | 4/4     | capping domain (856-1324)               |
| L       | 1293     | I->V                           | 4/4     | capping domain edge                     |
| L       | 1794     | S->I                           | 4/4     | MTase domain; hotspot; 10 aa from K1784 |

Table 6. Substitutions of the 2024 Gujarat isolate vs the 2003-2007 Indian human consensus (support = fraction of the four 2003-2007 genomes carrying the consensus residue).

### 3.7 Structure mapping: variable sites avoid catalytic centers and the encapsidated RNA

Of 270 variable L sites, 261 mapped onto 6U1X; of 57 variable N sites, all 57 mapped onto 2GIC. Variable L sites keep their distance from the RdRp active center (median ~50 Å from the GDN motif; Figure 4, left). The three tdCE causal sites map at 33.0 Å (L819, inter-domain), 38.4 Å (L978, capping) and 56.0 Å (L1658, MTase) from the GDN motif - consistent with allosteric or folding-mediated host-range effects rather than direct catalytic disruption [12]. In N, variable residues sit significantly farther from the encapsidated RNA than conserved residues (22.7 Å vs 17.4 Å; Mann-Whitney  $p=3.1\text{e-}05$ ; Figure 4, right): the RNA-binding surface of the nucleocapsid is protected, and variation is pushed to the solvent-exposed exterior.

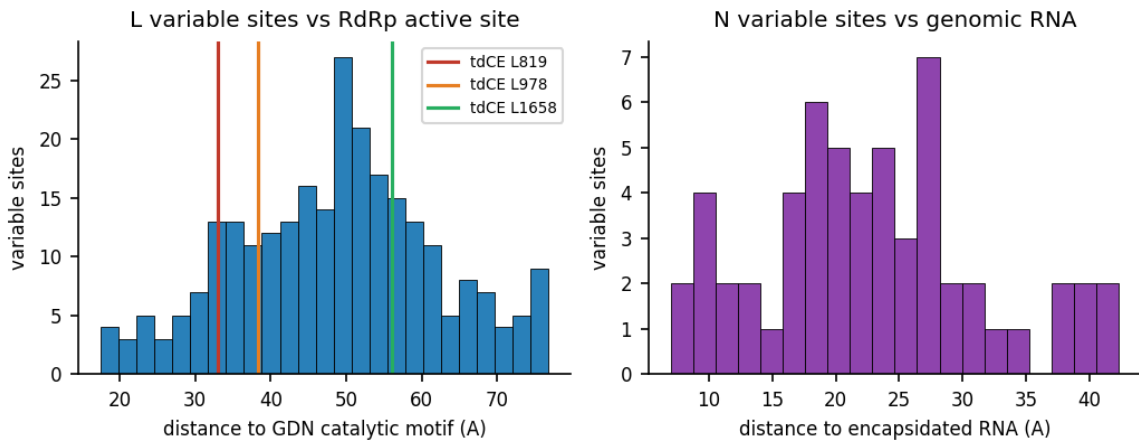


Figure 4. Structure-mapped variation. Left: distance of variable L sites from the GDN catalytic motif (VSV 6U1X mapping); tdCE causal sites marked. Right: distance of variable N sites from the encapsidated RNA (VSV 2GIC mapping).

### 3.8 Tool deliverable: chpv-profile

The complete workflow ships as a reusable CLI (chpv-profile, code/): fetch (NCBI eutils -> per-accession FASTA + md5 manifest), landmarks (gate G1: curated-landmark scanner + catalytic-invariance control + blinded tdCE known-answer recovery), spectrum (gate G2: per-site entropy, accession-provenanced substitution table, host stratification with Fisher tests, exact count reconciliation), structure (gate G3: template mapping with per-domain identity gating, distances to catalytic centers and RNA), figures (fig1-fig5, deterministic), paper (this PDF, generated from results JSONs - no number is typed by hand). The same gated, positive-control-first pattern transfers directly to the sibling Nipah slice of this repository.

## 4. Discussion

Three findings emerge. First, the CHPV replication machinery is under strong, domain-structured purifying selection: catalytic cores (RdRp, connector, HR motif, MTase tetrad, P Trp135) are invariant, and variation that does accumulate is excluded from the N oligomerization region and pushed away from the encapsidated RNA and from the RdRp active center. This mirrors the architecture-level conservation seen in VSV [16,17] and extends it to a human pathogen with 59 years of sampled evolution. Second, the deepest variation in the dataset is the India-Africa split, which we deliberately do not call host adaptation: human isolates are all Indian and vector isolates all African, so host and geography are inseparable with current sampling. Resolving this requires Indian vector isolates or African human isolates - neither exists in public data. Third, the human-transmitted lineage is remarkably static: the 2024 outbreak virus is only 17 L/P/N residues removed from the 1965 reference, implying that the 2024 Gujarat outbreak reflects ecology and surveillance rather than viral adaptation. This agrees with the declining case fatality rate being attributable to supportive care rather than attenuation [4].

Compared with prior art, this study adds the missing joint layer. The drug-repurposing study of 2026 [15] modelled CHPV structures but summarized genomes as per-gene average identities without per-site resolution, provenance, or host stratification; the phylogenetic study [6] resolved clades but not residue-level functional context; functional papers mapped individual landmarks [8,10,11,12] without a genomic backdrop. Here, landmarks, spectrum, host stratification and structure are integrated under pre-registered gates with positive controls - and the controls matter: the tdCE recovery (Table 1) demonstrates the pipeline can reproduce a known experimental answer exactly before any novel claim is made.

The tdCE structure mapping deserves emphasis because it constrains mechanism. All three causal host-range substitutions map tens of angstroms from the catalytic residues (33-56 A from GDN; 29-50 A from the HR histidine), in three different structural neighborhoods: L819 between the RdRp and capping domains, L978 inside the capping domain, and L1658 inside the MTase domain - the last immediately preceding the GDGSG motif highlighted by Stock et al. [12]. That three host-range determinants in one protein are all allosteric to the active centers suggests CHPV host range is tuned through domain-domain dynamics of L rather than through catalytic chemistry - consistent with the observation that catalytic chemistry itself is absolutely conserved (section 3.2). In the VSV L structure, these neighborhoods correspond to interfaces that rearrange during the initiation-to-elongation transition [16], which makes them plausible sensors of host-factor engagement.

The near-stasis of the human lineage also has a surveillance consequence. Because only ~17 L/P/N residues separate 1965 from 2024, single-genome outbreak reports (such as the 2024 Gujarat study [4]) cannot, by themselves, detect adaptation: any new human isolate will look almost identical to every previous one. Claims of increased virulence require functional assays, not sequence proximity. The substitution register in Table 6 and Appendix D provides the exact baseline against which future isolates should be scored.

**Limitations.** (i) Only 5 human genomes exist; Fisher p-values are descriptive, and the human-vector contrast is fully confounded with geography (section 3.5). (ii) Structure mapping uses VSV templates (51-61% identity), not CHPV experimental structures; claims are restricted to  $\geq 40\%$ -identity regions per the locked gate, but local geometry may differ in CHPV. (iii) AlphaFold DB lacks CHPV models; a future colabfold-scale prediction of selected CHPV targets (N oligomer, P CTD, L domains) would tighten the structural claims. (iv) The P analysis

treats the 310-aa hedgehog P via pairwise alignment; domain-level P claims are qualitative by design because P is largely disordered [9]. (v) Nucleotide-level selection analysis (dN/dS) was out of scope; amino-acid entropy is the variability metric here.

**Priority follow-ups.** (1) The 2024 isolate carries three substitutions in/adjacent to the L MTase domain (H1769N, S1794I, L1852F); cap-methylation assays or minigenome systems could test functional impact. (2) Indian sandfly sequences would break the host-geography confound. (3) The L1794 hotspot - independently substituted in three lineages (Glu in Senegal/Nigeria, Asp in Kenya, Ile in Gujarat 2024) - is a candidate for epistatic profiling. (4) Cross-disease transfer of this gated, positive-control-first pipeline to Nipah (the sibling slice in this repository).

## 5. Conclusions

Across 27 genomes and 59 years, the Chandipura virus replication machinery is conserved where it catalyses and variable where it does not: catalytic residues are invariant, catalytic domains are purged, and remaining variation avoids the RNA-binding surface and the polymerase active center. The dominant divergence is a deep India-Africa lineage split that current sampling cannot attribute to host adaptation, and the human outbreak lineage is near-static from 1965 to 2024. All claims carry accession-level provenance, all gates were locked before outcomes were examined, and the pipeline (chpv-profile) reproduces a published experimental mutation table exactly.

## 6. Data and code availability

Genomes: NCBI nuccore txid11272 (accessions in Appendix A). Templates: PDB 6U1X, 2GIC. Code, manifests, checksums, results JSONs, figures and this paper: [chandipura/](#) slice of the [chandipura-nipah-profiling](#) repository. Pipeline: [chpv-profile](#) (code/). Gates: `LOCKED_GATES.md`. Counts verified: substitution table rows (4,768) equal site-level sums exactly.

## Author contributions and declarations

U.P. designed the study, locked the gates, built and ran the pipeline, interpreted results and wrote the paper. Analysis was executed with the [chpv-profile](#) pipeline in a 2-core/2 GB sandbox on public data only. No funding was received for this work. The author declares no competing interests. This is a computational study of public sequences; no new biological materials were generated and no ethical approvals were required.

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## Appendix A. Accession manifest (28 fetched records; NC\_020805.1 excluded as identical RefSeq copy)

| Accession   | nt    | Host                   | Year        | Isolate               | MD5 (first 8) |
|-------------|-------|------------------------|-------------|-----------------------|---------------|
| GU190711.1  | 11094 | Homo sapiens           | 2007        | CIN 0728              | 124f346f      |
| GU212856.1  | 11120 | Homo sapiens           | 2004        | CIN 0451              | 3991572f      |
| GU212857.1  | 11083 | Homo sapiens           | 2007        | CIN 0755              | db56791a      |
| GU212858.1  | 11120 | Homo sapiens           | 2003        | CIN 0327              | 66e340ce      |
| HM627186.1  | 11061 | hedgehog               | 1966        | unknown               | cb8bec15      |
| HM627187.1  | 11067 | sandfly                | 1978        | unknown               | 8e187d7f      |
| KF468772.1  | 11119 | unknown                | unknown     | temperature-dependent | 48b507c8      |
| KF468773.1  | 11119 | unknown                | unknown     | temperature-dependent | 2c8bcc38      |
| KF468774.1  | 11119 | unknown                | unknown     | temperature-dependent | 895ef3de      |
| KF468775.1  | 11119 | unknown                | unknown     | I653514               | a208d148      |
| MT019608.1  | 11061 | Phlebotomine sandflies | 1995        | ArD111761             | 419c8d70      |
| MT019609.1  | 11061 | Phlebotomine sandflies | 1992        | ArD89384              | 129ff851      |
| MT019610.1  | 11061 | Phlebotomine sandflies | 1994        | ArD105134             | 9145c8a8      |
| MT019611.1  | 11061 | Phlebotomine sandflies | 1995        | ArD111758             | 891c5f4c      |
| MT019612.1  | 11061 | Phlebotomine sandflies | 1997        | ArD129185             | 0b5f4470      |
| MT019613.1  | 11061 | Phlebotomine sandflies | 1997        | ArD129199             | e0534af8      |
| MT019614.1  | 11061 | Phlebotomine sandflies | 1997        | ArD129212             | 074ddb8a      |
| MT019615.1  | 11061 | Phlebotomine sandflies | 1995        | ArD111757             | 3bfab785      |
| MT019616.1  | 11061 | Phlebotomine sandflies | 1997        | ArD129178             | 96eb2a7f      |
| MT019617.1  | 11077 | Phlebotomine sandflies | 1995        | ArD111213             | e3409108      |
| MT019618.1  | 11077 | Phlebotomine sandflies | 1995        | ArD111313             | db72e338      |
| MT019619.1  | 11077 | Phlebotomine sandflies | 1997        | ArD129179             | 2bb6db18      |
| NC_020805.1 | 11120 | Homo sapiens           | 2004        | CIN 0451              | 982befbd      |
| ON158116.1  | 11109 | phlebotomine sand flie | 13-Dec-2016 | BAR_SS10_763          | 2c15f679      |
| ON158117.1  | 11109 | phlebotomine sand flie | 25-Jun-2017 | BAR_SS12_700          | 6c81c765      |
| ON158118.1  | 11109 | phlebotomine sand flie | 20-Aug-2016 | TUR_SS1_361           | 4d42ff18      |
| ON158119.1  | 11117 | phlebotomine sand flie | 25-Jun-2017 | TUR_SS5_851           | c77f8862      |
| PQ185534.2  | 11078 | Homo sapiens           | 2024-07-20  | CHPV0744              | 39be2849      |

Full checksums: data/manifest.json and data/payload\_checksums.md5 in the repository.

## Appendix B. Locked gates (verbatim)

### CHAN-STRUCT: Chandipura virus L/P/N structural + mutation profiling

#### ## Research question

Do CHPV replication proteins (L polymerase, P phosphoprotein, N nucleoprotein) carry host-stratified (human vs vector/animal) amino-acid substitution patterns, and where do variable sites fall relative to functional domains and 3D structure?

#### ## Hypothesis

Human-derived isolates carry a distinct set of substitutions vs vector/animal isolates, concentrated OUTSIDE catalytic/binding cores (purifying selection on function); human-associated substitutions, if any, map to surface/peripheral regions of L/P/N structures.

#### ## Prior-art verdict: CROWDED (gap identified)

Closest works:

1. PINSA 2026, doi:10.1007/s43538-026-00906-8 - AlphaFold3 models of 5 CHPV proteins + drug docking; 29-genome per-gene average identity only. No per-site mutation spectrum, no host stratification, no accession provenance.
  2. IJMS 2025, doi:10.3390/ijms26031021 - phylogenetics of 23 genomes (human/sandfly/hedgehog); clades only, no per-residue spectrum, no structure.
  3. Microbiol Spectr 2025, doi:10.1128/spectrum.01578-25 - one 2024 Gujarat genome.
  4. PLoS ONE 2012, doi:10.1371/journal.pone.0030315 - whole-genome comparison.
  5. Ogino 2010, doi:10.1099/vir.0.019307-0 - HR motif / PRNTase biochemistry (L).
  6. tdCE host-range paper, doi:10.1099/vir.0.059204-0 - L mutations set host range.
- GAP: no study combines curated-domain positive-control validation, a per-site accession-provenanced mutation spectrum across ALL complete genomes stratified by host, and structure-mapped interpretation of that spectrum. That is the claim.

#### ## GATES

- G0 byte-lock: all CHPV genomes txid11272 SLEN 10000-12000 nt fetched from NCBI eutils; manifest of accession, host, country, year, md5 per file.

PASS: >=20 genomes total; >=5 non-human isolates (else host-stratified claims downgraded and reported thin).

- G1 positive control (landmark recovery): automated scan of reference L/P/N must recover curated knowns BEFORE any novel claim:

L: RdRp motifs A-F incl. catalytic GDN in motif C; HR capping motif (His-Arg);

C-terminal MTase catalytic tetrad region (K-D-K-E).

P: dimerization region around W135; N0-P binding region; disordered N-term.

N: N0-P interaction region (aa 1-180); N-RNA-P region (aa 320-390).

PASS: >=80% of curated landmarks recovered within +/-20 aa of curated position.

- G2 mutation spectrum: per-site amino-acid variability for L, P, N across all genomes; every substitution traceable to accession(s) in a provenance table; human-vs-animal contingency with Fisher exact (any cell <5); verified counts (script totals == table totals, printed in report).

PASS: 100% of reported substitutions have accession-level provenance; totals reconcile exactly.

- G3 structure mapping: variable/conserved sites mapped onto structure.

AlphaFold DB model for N (P11211) if pLDDT>=70 in mapped regions; L (2092 aa) via VSV L template PDB 6U1X homology mapping if no full-length AF model meets quality bar; P treated as partly disordered (Sci Rep 2021 disorder analysis), mapped regions restricted accordingly.

PASS: structural claims only over validated regions (pLDDT>=70 or template identity >=40%); quality metrics printed per mapped region.

- G4 reproducibility: figures regenerate from manifest+code; md5 manifest covers every payload file; README reproduces end-to-end in fresh sandbox.

#### ## Negative-result protocol

Null findings (e.g., no host stratification) are reported as informative negatives with power statement, never re-fished.

#### ## Tool deliverable

Reusable CLI: chpv-profile (download -> align -> per-site spectrum -> provenance

-> structure map -> figures), packaged in repo with README.



## Appendix C. Provenance excerpt (first 50 of 4,768 substitution records)

| Gene | Pos | Ref | Alt | Accession  | Group    |
|------|-----|-----|-----|------------|----------|
| N    | 2   | S   | N   | ON158118.1 | vector   |
| N    | 2   | S   | N   | ON158116.1 | vector   |
| N    | 4   | Q   | K   | ON158119.1 | vector   |
| N    | 4   | Q   | K   | ON158118.1 | vector   |
| N    | 4   | Q   | K   | ON158117.1 | vector   |
| N    | 4   | Q   | K   | ON158116.1 | vector   |
| N    | 4   | Q   | K   | MT019619.1 | vector   |
| N    | 4   | Q   | K   | MT019618.1 | vector   |
| N    | 4   | Q   | K   | MT019617.1 | vector   |
| N    | 4   | Q   | K   | MT019616.1 | vector   |
| N    | 4   | Q   | K   | MT019615.1 | vector   |
| N    | 4   | Q   | K   | MT019614.1 | vector   |
| N    | 4   | Q   | K   | MT019613.1 | vector   |
| N    | 4   | Q   | K   | MT019612.1 | vector   |
| N    | 4   | Q   | K   | MT019611.1 | vector   |
| N    | 4   | Q   | K   | MT019610.1 | vector   |
| N    | 4   | Q   | K   | MT019609.1 | vector   |
| N    | 4   | Q   | K   | MT019608.1 | vector   |
| N    | 4   | Q   | K   | HM627187.1 | vector   |
| N    | 4   | Q   | K   | HM627186.1 | hedgehog |
| N    | 6   | F   | L   | MT019619.1 | vector   |
| N    | 6   | F   | L   | MT019616.1 | vector   |
| N    | 6   | F   | L   | MT019615.1 | vector   |
| N    | 6   | F   | L   | MT019614.1 | vector   |
| N    | 6   | F   | L   | MT019613.1 | vector   |
| N    | 6   | F   | L   | MT019612.1 | vector   |
| N    | 6   | F   | L   | MT019611.1 | vector   |
| N    | 6   | F   | L   | MT019610.1 | vector   |
| N    | 6   | F   | L   | MT019609.1 | vector   |
| N    | 6   | F   | L   | MT019608.1 | vector   |
| N    | 6   | F   | L   | HM627186.1 | hedgehog |
| N    | 9   | S   | A   | ON158119.1 | vector   |
| N    | 9   | S   | A   | ON158118.1 | vector   |
| N    | 9   | S   | A   | ON158117.1 | vector   |
| N    | 9   | S   | A   | ON158116.1 | vector   |
| N    | 9   | S   | A   | MT019619.1 | vector   |
| N    | 9   | S   | A   | MT019618.1 | vector   |
| N    | 9   | S   | A   | MT019617.1 | vector   |
| N    | 9   | S   | A   | MT019616.1 | vector   |

| Gene | Pos | Ref | Alt | Accession  | Group    |
|------|-----|-----|-----|------------|----------|
| N    | 9   | S   | A   | MT019615.1 | vector   |
| N    | 9   | S   | A   | MT019614.1 | vector   |
| N    | 9   | S   | A   | MT019613.1 | vector   |
| N    | 9   | S   | A   | MT019612.1 | vector   |
| N    | 9   | S   | A   | MT019611.1 | vector   |
| N    | 9   | S   | A   | MT019610.1 | vector   |
| N    | 9   | S   | A   | MT019609.1 | vector   |
| N    | 9   | S   | A   | MT019608.1 | vector   |
| N    | 9   | S   | A   | HM627187.1 | vector   |
| N    | 9   | S   | A   | HM627186.1 | hedgehog |
| N    | 29  | G   | S   | ON158119.1 | vector   |

Complete table: results/g2\_spectrum.json (subs\_table), reconciled exactly against site-level sums.

## Appendix D. Complete substitution register: 2024 Gujarat isolate (PQ185534.2) vs 2003-2007 consensus

| Protein | Position | Change | Support | Region / note                     |
|---------|----------|--------|---------|-----------------------------------|
| N       | 163      | A->T   | 4/4     | N0-P interaction region (1-180)   |
| P       | 52       | E->A   | 4/4     | between disordered regions 1-2    |
| P       | 112      | E->G   | 3/4     | reversion to 1965 reference state |
| L       | 188      | K->R   | 4/4     | N-terminal region                 |
| L       | 248      | S->P   | 4/4     | N-terminal region                 |
| L       | 317      | A->V   | 3/4     | N-terminal region                 |
| L       | 375      | N->D   | 4/4     | N-terminal region                 |
| L       | 1062     | R->K   | 4/4     | capping domain (856-1324)         |
| L       | 1293     | I->V   | 4/4     | capping domain edge               |
| L       | 1794     | S->I   | 4/4     | MTase domain; hotspot (see 3.5.2) |
| L       | 2050     | S->N   | 4/4     | C-terminal region                 |

11 substitutions vs the 2003-2007 consensus. Against the 1965 reference directly the 2024 isolate differs at 17 L/P/N residues (N 2, P 3, L 12); full per-protein diff in results/g5\_temporal.json.

## Appendix E. Verified/thin/missing statement

**Verified:** all substitution counts (reconciled 4,768 = 4,768); all landmark recoveries (12/12); catalytic invariance (10/10 residues x 27 genomes); tdCE recovery (7/7 coding, 3/3 silent); template identities (computed from sequences extracted from the mmCIF files used for mapping); all figure numbers (regenerated from results JSONs). **Thin:** human-vs-vector statistics (n=5 vs n=17, fully confounded with geography - treated descriptively); P structure claims (protein is largely disordered; qualitative only). **Missing:** no CHPV experimental structure for L, P, or N (mapping is on VSV templates); no AlphaFold DB CHPV models; no Indian vector or African human isolates in public data; no dN/dS analysis (out of scope); P length outlier (HM627186.1, 310 aa) handled by pairwise alignment, domain-level P claims restricted accordingly.